

Lesson 1.3: Analyzing Interview Data

Location: Classroom

Grade Level: High School

Time: 50 min

Driving Question

- What values do community members place on the ocean?

Lesson Synopsis

In this 50-minute lesson, students analyze the notes they take during the interviews to identify key themes and patterns in how people in their community connect with the ocean.

They begin by reviewing their notes to categorize responses into three main areas: past experiences with the ocean, reasons they care about the ocean, and perspectives on how the ocean is changing.

Using thematic posters, students sort and organize ideas to visualize commonalities across interviews. As a class, they then discuss emerging patterns, exploring the shared values, concerns, and lived experiences that shape their community's relationship with the ocean. Finally, students consider how these insights might inform their own understanding of ocean change and its impact on their community.

What Students Will Figure Out:

- Engaging in a process of data analysis can uncover key patterns and themes in our community's relationship with the ocean, examining past experiences, personal values, and perceptions of ocean change.
- By categorizing and visualizing interview responses, students can identify shared concerns and values that reflect the community's collective connection to the ocean.

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Analyze their interview notes to categorize responses into key themes.	Class posters created during the <i>Investigate</i> section.
2. Identify commonalities and differences in interview responses , discussing patterns in how their community values and perceives the ocean.	Class discussion during the <i>Navigate</i> section.
3. Identify key ideas from their analysis that could be incorporated into a model of how the ocean is changing.	Class discussion during the <i>Navigate</i> section.

Building towards NGSS and AI Literacy:

- **HS-ESS3-1:** Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

LESSON OVERVIEW

Section	Time	What Happens
Navigate	10 minutes	Students share their initial reflections on their experience with the interviews, including what they noticed and what stood out to them.
Investigate	30 minutes	Students review their interview notes to identify their interviewees' past experiences with the ocean, reasons they care about the ocean, and perspectives on how the ocean is changing. After recording these ideas on sticky notes, students sort them onto thematic posters around the room so that they can look for patterns across the entire group.
Putting pieces together as a class	15 minutes	Students take part in a gallery walk to look for patterns and themes. As a whole class, they then discuss commonalities and differences in the interview responses.
Navigate	5 minutes	Students break back into their teams to reflect on the patterns that they noticed and identify key ideas that they might incorporate into their models.

MATERIALS

- [Module 1.3 Slideshow](#)
- [Science Journal](#)
- A screen to view the slideshow

SUPPLIES

- Student notes from their interviews
- (2) highlighters of different colors (i.e., yellow, blue, pink) per student team
- (3) sets of post-it notes with matching colors (i.e., yellow, blue, pink) per student team (optional)
- Poster boards or large pieces of paper (optional)
- Colored markers (optional)

BEFORE YOU START TEACHING

1. Decide whether you will have students sort the interview themes onto a physical poster board or onto a virtual document.

If you choose to physically sort, make sure to set up the poster boards for each in advance. Color-code each theme using colored markers, and then physically write the name of each theme on the poster board.

2. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: In this lesson, students have collected interviews from their communities and analyzed the themes in these interviews. Students are introduced to qualitative data analysis of interview data and organize their findings in thematic posters. By the end of the lesson, students should have a set of initial themes that they will focus on to construct their initial model of climate change's impact on Southern California's ocean in Module 2. The lesson will cover:

HS-ESS3-1: Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

Students will encounter the terms: thematic analysis, patterns, and community values. They might revisit interview etiquette when sharing about their interview experiences. They will continue to think about ideas related to climate change and how our ocean is changing, such as rising sea levels, rising temperatures, fish populations, and biodiversity.

Where We Are NOT Going: The lesson focuses on analyzing and reflecting on themes from the

community interviews, instead of conducting additional research on how Southern California's ocean is changing. Students might mention rising sea levels, coastal erosion, and biodiversity loss for future investigations, but we will not investigate these phenomena in depth yet.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: Interview Reflection (10 minutes)

1. Open the slideshow to **Slide 1** and introduce lesson 1.3.
2. Advance the slideshow to **Slide 2**, and as a class, ask students to share their team's experience with conducting an interview.
 - Who did you interview?
 - What was your experience with the interviews?
 - What did you notice? What stood out to you?
 - What questions do you still have?
3. Advance the slideshow to **Slide 3** and give students an overview of the day. Inform students that now that they have conducted interviews, their next step is to analyze their interview data to make sense of it. Some goals of Lesson 1.3 will be to:
 - Analyze your interview notes to categorize them into themes.
 - Identify patterns in the interview notes to explore what values your community members place on the ocean.

INVESTIGATE: Interview Analysis (30 minutes)

1. Tell students that now that we've had a chance to interview community members to learn more about their experiences with the ocean, our next step is to analyze our interview notes to look for patterns.

Move on to **Slide 4** and invite students to share their initial ideas as a class:

- Why did you conduct a community interview?
- What information did you want to gather from the community interview?
- How might you analyze your interview notes to find out how climate change is affecting Southern California's ocean?

Give students a chance to share their initial ideas out loud.

2. Once students have shared their initial ideas, inform them that now that they have collected interview data, they need to analyze the responses to see why community members might care about the ocean.
3. Advance to **Slide 5**, tell students that following an interview, social scientists often have to make sense of the data. One way that they do this is by categorizing the responses of the interviews into categories of themes.

Share with students that they will work with their teams to categorize their data into themes, just like social scientists do. However, they will have three categories

- **Category #1:** Past experiences with the ocean
- **Category #2:** Reasons the interviewee cares about the ocean
- **Category #3:** Ideas about how the ocean is changing

Inform students that before they work on their interview notes, they will have a chance to practice with some examples.

4. Advance to **Slides 6-8** where students will have examples of interview responses to categorize. As a class, work together to identify which category each response will fit under, if it fits under more than one.
 - **Example 1:** “I grew up visiting the state parks and going to tidepools with my siblings.”
 - **Category #1: Past experiences with the ocean**
 - **Category #2: Reasons the interviewee cares about the ocean**
 - **Category #3: Ideas about how the ocean is changing**
 - **Example 2:** “We used to not have as many intense storms.”
 - **Category #1: Past experiences with the ocean**
 - **Category #2: Reasons the interviewee cares about the ocean**
 - **Category #3: Ideas about how the ocean is changing**
 - **Example 3:** “Food is the first thing that I think of. I love seafood, particularly octopus salad.”
 - **Category #1: Past experiences with the ocean**
 - **Category #2: Reasons the interviewee cares about the ocean**
 - **Category #3: Ideas about how the ocean is changing**

5. Once the class has had a chance to practice with the examples, advance to **Slide 9**. Have the students work with their teams to review their community interview notes (refer to **Module 1.2 Part 2: Conducting Community Interviews (Student Task: Optional Interview Notes)** in the Science Journal) to highlight or underline any ideas related to the three categories. If something does not fit into one of the three categories, they should not highlight it.

As students do so, move between the groups to answer questions and provide support on questions they may have.

6. After student teams have had an opportunity to highlight their ideas, tell them that their next step is to categorize these ideas.
7. Advance to **Slide 10** and ask students to create three category piles and place their post-it notes under the appropriate category.
 - **Category #1:** Past experiences with the ocean
 - **Category #2:** Reasons the interviewee cares about the ocean
 - **Category #3:** Ideas about how the ocean is changing

Note: If you have three different color post-it notes, it might be helpful to have students use a different post it note color for each category.

8. Once students have finished categorizing their responses from their community interview, advance to **Slide 11**. Tell them that our next step is to look for patterns in the responses that our interviewees shared and place these ideas on a poster around the classroom.
9. To do this, we'll sort the post-its into themes. These themes can be centered around values, or why an individual may feel connected to the ocean any may include:
 - Emotional
 - Wellbeing & feelings
 - Financial
 - Tradition

PUTTING PIECES TOGETHER AS A CLASS: Gallery Walk (15 minutes)

1. When student posters are ready, place them around the classroom and advance to **Slide 12**. Tell students that they will be participating in a gallery walk to see how other community members place value on the ocean.
2. Inform students that they will first conduct a gallery walk and then return as a class to discuss their findings. To structure the gallery walk, provide students the following questions to get them started.
 - What patterns or themes did you identify that are similar than the patterns or themes your team identified?
 - What patterns or themes did you identify that are different than the patterns or themes your team identified?
 - What changes might you make to your own poster after seeing other posters?
3. After the gallery walk, bring the class back together and advance to **Slide 13**. As a whole class, discuss the following:
 - Are there any common themes you noticed during the gallery walk?
 - What is a theme or pattern that you found interesting or different from your own?
 - How might you use these patterns to identify why the community cares about the ocean?

NAVIGATE: Lesson Recap (5 minutes)

1. As a final task, advance to **Slide 14** and give students some time to reflect on **Module 1.3: Analyzing Interview Data (Student Reflection), page 8** of their **Science Journals**.
 - What did you learn from analyzing your interview responses? How has your thinking about why people value the ocean or how it is changing been affected?
 - Why is it important to analyze responses to interview questions? What can this tell us about how communities value the ocean?
 - How will you use what you learned in Module 1 to build your model of how climate change is impacting Southern California's ocean? For example, what information will you add or show to explain these impacts?
2. Advance to **Slide 15**. As the time wraps up for the day, thank students for their time. Inform students that this concludes module 1. In module 2, students will step into the next step in the process, using their information to create a model.